Project: Worked examples
Title: Bolts and fillet welds (EN 1993-1-8)
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## 1. Scope and method

The design resistances of one bolt in a plate (shear, bearing, tension, punching, combined shear and tension), an optional block-tearing check of the bolt group, the spacing limits, and a fillet weld by the simplified and the directional method, each against its design force.
Bolt strengths: f_ub = class × 100 MPa; $A_s = \pi/4\cdot((d_2 + d_3)/2)^2$, $d_2 = d - 0.6495P$, $d_3 = d - 1.2269P$ Shear (cl. 3.6, Table 3.4): $F_{v,Rd} = \alpha_v \cdot f_{ub} \cdot A \cdot n_s/\gamma_{M2}$; $\alpha_v = 0.6$ (4.6, 5.6, 8.8 or shank in the plane), 0.5 (4.8, 5.8, 6.8, 10.9 with threads in the plane) Bearing: $F_{b,Rd} = k_1 \cdot \alpha_b \cdot f_u \cdot d \cdot t/\gamma_{M2}$; $\alpha_b = \min(\alpha_d, f_{ub}/f_u, 1)$, $\alpha_d = e_1/3d_0$ (end) or $p_1/3d_0 - 1/4$ (inner); $k_1 = \min(2.8e_2/d_0 - 1.7, 1.4p_2/d_0 - 1.7, 2.5)$ Tension: $F_{t,Rd} = k_2 \cdot f_{ub} \cdot A_s/\gamma_{M2}$, $k_2 = 0.9$; punching $B_{p,Rd} = 0.6\pi \cdot d_m \cdot t \cdot f_u/\gamma_{M2}$; combined $F_v/F_{v,Rd} + F_t/(1.4F_{t,Rd}) \le 1$ Block tearing (cl. 3.10.2): $V_{eff,1,Rd} = f_u \cdot A_{nt}/\gamma_{M2} + f_y \cdot A_{nv}/(\sqrt{3} \cdot \gamma_{M0})$ Weld, simplified (cl. 4.5.3.3): $f_{vw,d} = f_u/(\sqrt{3} \cdot \beta_w \cdot \gamma_{M2})$, $F_{w,Rd} = f_{vw,d} \cdot a$; directional (cl. 4.5.3.2): $\sqrt{(\sigma_{\perp}^2 + 3(\tau_{\perp}^2 + \tau_{\parallel}^2))} \le f_u/(\beta_w \cdot \gamma_{M2})$, $\sigma_{\perp} \le 0.9f_u/\gamma_{M2}$
Ref: EN 1993-1-8:2005 cl. 3.5, Table 3.3 — Positioning of holes for bolts
Ref: EN 1993-1-8:2005 cl. 3.6.1, Table 3.4 — Design resistance of individual fasteners
Ref: EN 1993-1-8:2005 cl. 3.10.2 — Block tearing
Ref: EN 1993-1-8:2005 cl. 4.5.3.2, 4.5.3.3, Table 4.1 — Fillet welds: directional and simplified methods; correlation factor
Ref: ISO 898-1:2013 cl. 9.1 — Nominal stress area (formula)

## 2. Inputs

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| grade | Bolt property class | 8.8 | | drawing: SPL-001 rev B | any |
| size | Bolt size | M20 | | drawing: SPL-001 rev B | any |
| n_s | Shear planes | 2 | | drawing: SPL-001 rev B | 1 … 4 |
| threads | Threads in the shear plane | yes | | assumption: Threads may lie in the shear planes (short bolts). | any |
| steel | Plate steel grade (β_w) | S275 | | drawing: SPL-001 rev B | any |
| t | Plate thickness | 10 | mm | drawing: SPL-001 rev B | 3 … 50 mm |
| f_y | Plate yield strength | 275 | MPa | spec: EN 10025-2 S275, t ≤ 16 mm | 235 … 460 MPa |
| f_u | Plate ultimate strength | 410 | MPa | spec: EN 10025-2 S275, 3 ≤ t ≤ 100 mm | 340 … 720 MPa |
| e_1 | End distance (load direction) | 40 | mm | drawing: SPL-001 rev B | 10 … 500 mm |
| e_2 | Edge distance (across the load) | 35 | mm | drawing: SPL-001 rev B | 10 … 500 mm |

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| p_1 | Pitch (load direction; blank: one row) | 70 | mm | drawing: SPL-001 rev B | 20 … 1000 mm |
| k_e1 | Minimum $e_1/d_o$ | 1.2 | | spec: EN 1993-1-8 Table 3.3 | 1 … 3 |
| k_e2 | Minimum $e_2/d_o$ | 1.2 | | spec: EN 1993-1-8 Table 3.3 | 1 … 3 |
| k_p1 | Minimum $p_1/d_o$ | 2.2 | | spec: EN 1993-1-8 Table 3.3 | 2 … 4 |
| k_p2 | Minimum $p_2/d_o$ | 2.4 | | spec: EN 1993-1-8 Table 3.3 | 2 … 4 |
| gamma_M2 | γ_M2 | 1.25 | | spec: UK NA to BS EN 1993-1-8, NA.2.3 | 1.1 … 1.5 |
| gamma_M0 | γ_M0 | 1 | | spec: UK NA to BS EN 1993-1-1, NA.2.15 | 1 … 1.2 |
| F_v_Ed | Shear per bolt | 55 | kN | assumption: Governing bolt from the bolt-group calculation (M3). | 0 … 2000 kN |
| F_t_Ed | Tension per bolt | 0 | kN | assumption: No tension. | 0 … 2000 kN |

## 3. Assumptions

A1. Static loading; category A (bearing type) shear and category D (non-preloaded) tension; no slip-resistant joints or moment end plates. (Scope of S4.)
A2. Bearing is taken at the worst of an end bolt and an inner bolt, and at an edge bolt across the load ($k_1$ with $e_2$ and $p_2$). (Conservative for every bolt in the group.)
A3. Spacing minima (Table 3.3) and γ_M factors are entered from a National Annex record; the values shown default to EN 1993-1-8 recommended. (§A5 small factor set.)
A4. Group forces come from M3 and weld forces per length from M6 by link. (§A6 S4 links.)
A5. threads (Threads in the shear plane) is taken as yes. (Threads may lie in the shear planes (short bolts).)
A6. F_v_Ed (Shear per bolt) is taken as 55 kN. (Governing bolt from the bolt-group calculation (M3).)
A7. F_t_Ed (Tension per bolt) is taken as 0 kN. (No tension.)

## 4. Working

[19] $f_{ub}$ = class first figure × 100 — EN 1993-1-8 Table 3.1
= class 8.8
= 800 MPa
Bolt ultimate strength

[20] $f_{yb} = f_{ub} \times$ second figure / 10 — EN 1993-1-8 Table 3.1
= class 8.8
= 640 MPa
Bolt yield strength

[21] $A_s = \pi/4 \cdot ((d_2 + d_3)/2)^2$, $d_2 = d - 0.6495P$, $d_3 = d - 1.2269P$ — ISO 898-1 cl. 9.1
= d = 20, P = 2.5 mm
= 244.79 mm²
Tensile stress area

[22] d_0 = d + 2 (≤ M24), d + 3 (≥ M27) unless entered
= 22 mm
= 22 mm
Hole diameter
[23] alpha_v = □_v (threads in the shear plane) EN 1993-1-8 Table 3.4
= class 8.8
= 0.6
Shear factor
[24] F_v_Rd = □_v·f_ub·A·n_s/□_M2 EN 1993-1-8 Table 3.4
= 0.6 × 800 × 244.79 × 2 / 1.25
= 188 kN
Shear resistance per bolt
[25] alpha_b = min(□_d, f_ub/f_u, 1), □_d = min($e_1/3d_0$, $p_1/3d_0$ - 1/4) EN 1993-1-8 Table 3.4
= min(0.60606, 0.81061, 1.9512, 1)
= 0.60606
Bearing factor α_b
[26] k_1 = min($2.8e_2/d_0$ - 1.7, $1.4p_2/d_0$ - 1.7, 2.5) EN 1993-1-8 Table 3.4
= $e_2$ = 35, $p_2$ = —, $d_0$ = 22
= 2.5
Bearing factor $k_1$
[27] F_b_Rd = $k_1$·□_b·f_u·d·t/□_M2 EN 1993-1-8 Table 3.4
= 2.5 × 0.60606 × 410 × 20 × 10 / 1.25
= 99.394 kN
Bearing resistance per bolt
[28] F_t_Rd = $k_2$·f_ub·A_s/□_M2, $k_2$ = 0.9 EN 1993-1-8 Table 3.4
= 0.9 × 800 × 244.79 / 1.25
= 141 kN
Tension resistance per bolt
[29] B_p_Rd = 0.6π·d_m·t·f_u/□_M2, d_m = (s + e)/2 EN 1993-1-8 Table 3.4
= 0.6π × 32.321 × 10 × 410 / 1.25
= 199.83 kN
Punching shear resistance
[30] u_v = F_v_Ed / F_v_Rd
= 55 kN / 188 kN
= 0.29255
Shear utilisation
[31] u_b = F_v_Ed / F_b_Rd
= 55 kN / 99.394 kN
= 0.55335
Bearing utilisation
[32] u_t = F_t_Ed / min(F_t_Rd, B_p_Rd)
= 0 kN / min(141 kN, 199.83 kN)
= 0
Tension utilisation (tension or punching)
[33] u_vt = F_v_Ed / F_v_Rd + F_t_Ed / (1.4 * F_t_Rd) EN 1993-1-8 Table 3.4
= 55 kN / 188 kN + 0 kN / (1.4 × 141 kN)
= 0.29255
Combined shear and tension

## 5. Results

| SYMBOL | VALUE | DESCRIPTION |
|---|---|---|
| f_ub | 800 MPa | Bolt ultimate strength |
| A_s | 244.79 mm² | Stress area |

| SYMBOL | VALUE | DESCRIPTION |
|---|---|---|
| alpha_v | 0.6 | Shear factor |
| F_v_Rd | 188 kN | Shear resistance |
| alpha_b | 0.60606 | Bearing factor |
| k_1 | 2.5 | Bearing factor $k_1$ |
| F_b_Rd | 99.394 kN | Bearing resistance |
| F_t_Rd | 141 kN | Tension resistance |
| B_p_Rd | 199.83 kN | Punching resistance |
| u_v | 0.29255 | Shear utilisation |
| u_b | 0.55335 | Bearing utilisation |
| u_t | 0 | Tension utilisation |
| u_vt | 0.29255 | Combined utilisation |
| governing | "bearing (0.553)" | Governing bolt mode |

## 6. Checks (governing first)

| RESULT | CHECK | DEMAND | CAPACITY | UTILISATION | REF |
|---|---|---|---|---|---|
| PASS | Bolt bearing | u_b = 0.55335 | u_b ≤ 1 | 0.553 | EN 1993-1-8 Table 3.4 |
| PASS | Bolt shear | u_v = 0.29255 | u_v ≤ 1 | 0.293 | EN 1993-1-8 Table 3.4 |
| PASS | Combined shear and tension | u_vt = 0.29255 | u_vt ≤ 1 | 0.293 | EN 1993-1-8 Table 3.4 |
| PASS | Bolt tension and punching | u_t = 0 | u_t ≤ 1 | 0 | EN 1993-1-8 Table 3.4 |
| PASS | Spacing and edge distances (Table 3.3 minima) | $e_1$ 40, $e_2$ 35, $p_1$ ≥ 26.4, 26.4, $p_2$ — mm 70, 48.4, 52.8 mm | | 0 | EN 1993-1-8 Table 3.3 |

## 7. Applicability

WARNING — record.gamma_M2: gamma_M2: a tabulated value not taken from a reference record (EN 1993-1-8 National Annex factors); enter it as one so its source, edition and licence travel with it.
WARNING — record.gamma_M0: gamma_M0: a tabulated value not taken from a reference record (EN 1993-1-8 National Annex factors); enter it as one so its source, edition and licence travel with it.
WARNING — record.k_e1: k_e1: a tabulated value not taken from a reference record (EN 1993-1-8 National Annex factors); enter it as one so its source, edition and licence travel with it.
WARNING — record.k_e2: k_e2: a tabulated value not taken from a reference record (EN 1993-1-8 National Annex factors); enter it as one so its source, edition and licence travel with it.
WARNING — record.k_p1: k_p1: a tabulated value not taken from a reference record (EN 1993-1-8 National Annex factors); enter it as one so its source, edition and licence travel with it.
WARNING — record.k_p2: k_p2: a tabulated value not taken from a reference record (EN 1993-1-8 National Annex factors); enter it as one so its source, edition and licence travel with it.
WARNING — record.f_y: f_y: a tabulated value not taken from a reference record (Plate strengths); enter it as one so its source, edition and licence travel with it.
WARNING — record.f_u: f_u: a tabulated value not taken from a reference record (Plate strengths); enter it as one so its source, edition and licence travel with it.

## 8. Verification

Template struct.connection-ec3@1.0.0 Content hash 2b3e26b3b4a4a10f Engine calc-engine/1.0.0
Known-answer cases passed 5/5 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0